

POLITECNICO
MILANO 1863

Requirement Analysis and Specification Document, Luca Suzani, Alberto Spezzapria, Alessio Pesavento

# Requirement Analysis and Specification Document

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# Chapter 1

# Introduction

## 1.1 Purpose

The purpose of the *Best Bike Paths (BBP)* system is to promote cycling as a sustainable, safe, and conscious means of transportation by providing cyclists with a digital platform for collecting, accessing, and sharing information about cycling routes and activities.
The system addresses the need for reliable and up-to-date data on bike paths, allowing users to track their performance while contributing to the improvement of shared information.
Through active community participation, BBP aims to identify and maintain the safest and most efficient routes, fostering a more informed and widespread cycling mobility.
Ultimately, BBP seeks to enhance the overall cycling experience by integrating multiple functionalities and creating a collaborative, continuously updated information network driven by user contributions.

### 1.1.1 Goals

G1. **Support cyclists in tracking their activities:** Enable users to record their cycling routes and access detailed statistics such as total distance, average speed, and other performance indicators, enriched—when available—by meteorological data from external services (e.g., weather conditions, temperature, and wind speed).

G2. **Allow cyclists to contribute information about bike paths:** Provide two modes for entering information related to bike paths:

- **Manual mode:** the user manually inputs the path data and its condition.
- **Automatic mode:** the system automatically collects data via GPS, accelerometer, and gyroscope during the cycling activity, requiring user confirmation before publication.

G3. **Enable the publication of recorded path information:** Allow users to make their recorded path information public, contributing to a shared and valuable database for the entire cycling community.

G4. **Enable route search and visualization:** Allow users to search for and display on a map one or more cycling routes connecting a chosen origin and destination, ranking them based on overall condition and effectiveness in reaching the goal.

G5. **Maintain up-to-date and reliable path information:** Ensure continuous updating of path data through the aggregation and analysis of user feedback, taking into account data freshness, consistency among reports, and user confirmations.

## 1.2 Scope

### 1.2.1 World Phenomena

### 1.2.2 Shared Phenomena

## 1.3 Definitions, Acronyms, and Abbreviations

## 1.4 Revision History

## 1.5 Reference Documents

## 1.6 Document Structure

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# Chapter 2

# Overall Description

Here you can see how to include an image in your document.

Here is the command to refer to another element (section, figure, table, ...) in the document: *As discussed in Section 2 and as shown in Figure 2.1, ....* Here is how to introduce a bibliographic citation [**?**]. Bibliographic references should be included in a `.bib` file.

Table generation is a bit complicated in Latex. You will soon become proficient, but to start you can rely on tools or external services. See for instance this https://www.tablesgenerator.com.

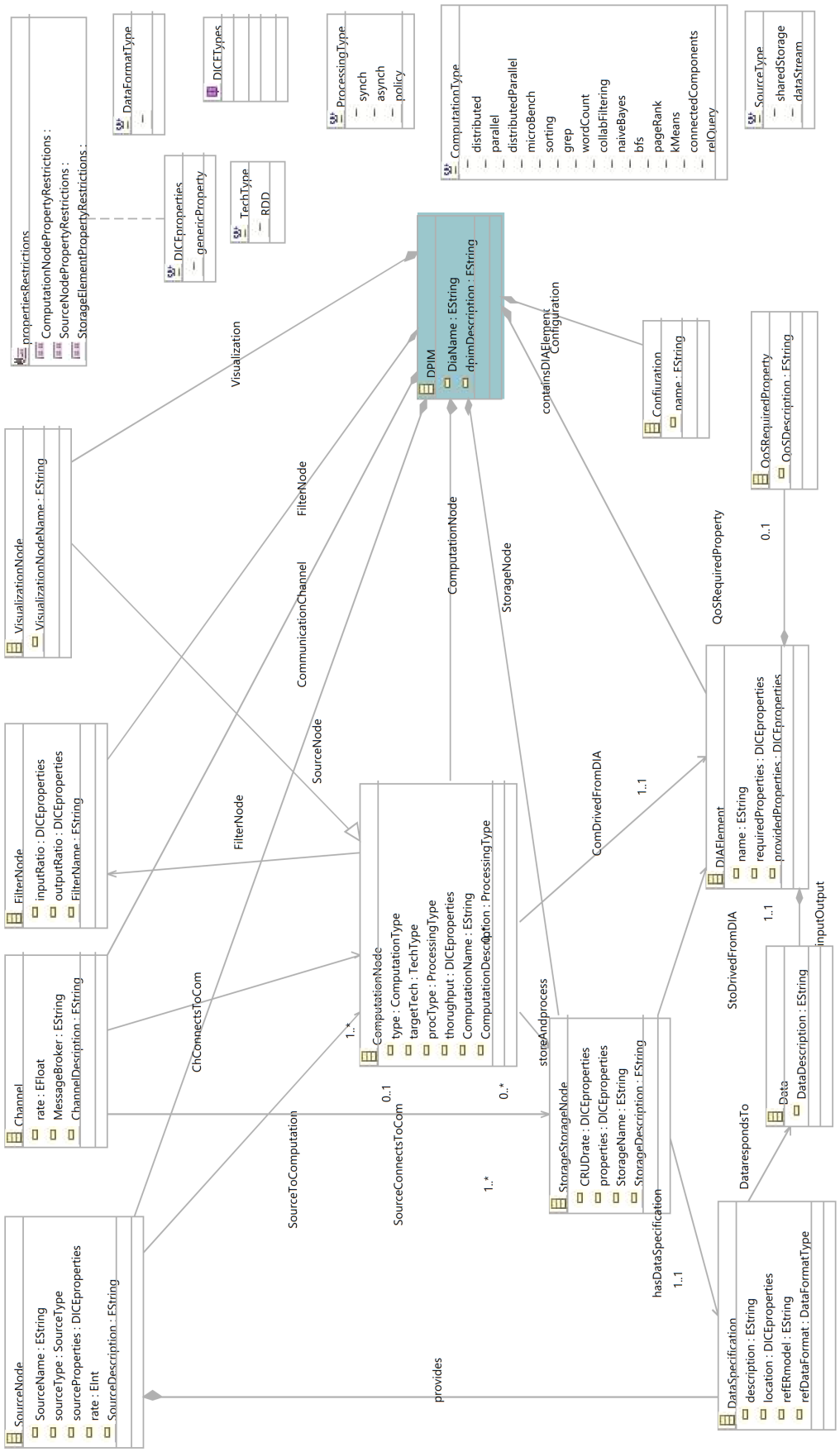



Figure 2.1: DICE DPIM metamodel.

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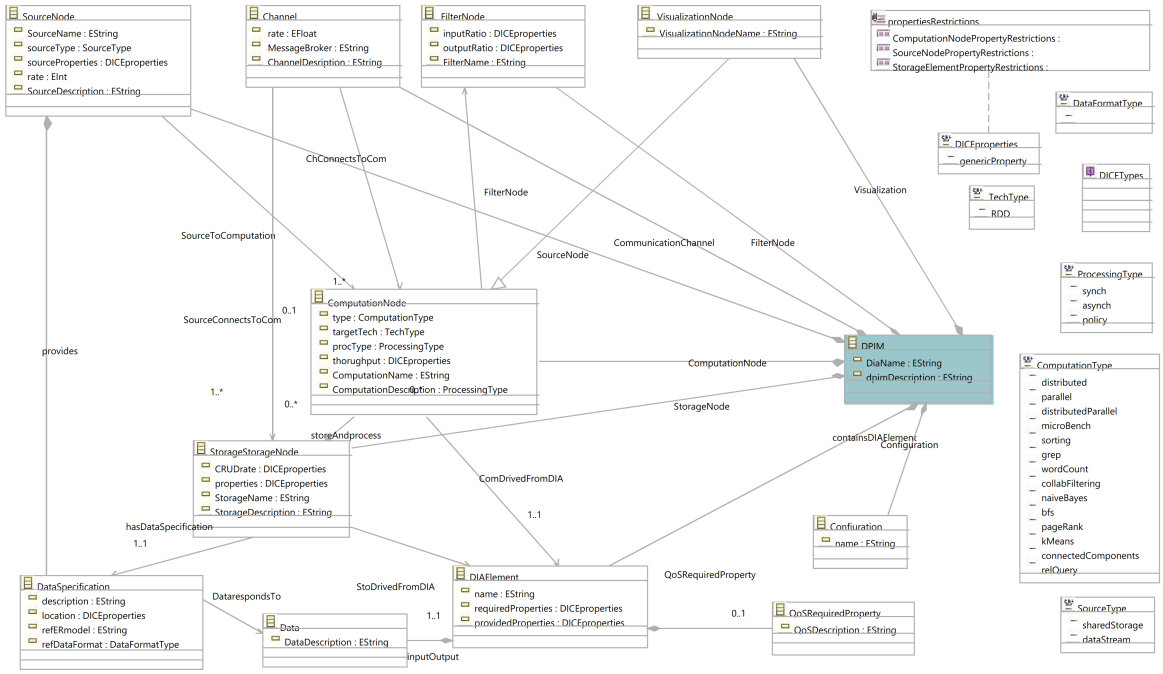



Figure 2.2: DICE DPIM metamodel in portrait form.

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# Chapter 3

# Specific Requirements

Organize this section according to the rules defined in the project description.

# Chapter 4

# Formal Analysis Using Alloy

Organize this section according to the rules defined in the project description.

# Chapter 5

# Effort Spent

| Data | Alessio Pesavento | Alberto Spezzapria | Luca Suzani |
|---|---|---|---|
| 27/10/2025 | 1h | 1h | 1h |
| 07/11/2025 | 2h | 2h | 2h |
| 08/11/2025 | - | 1h | - |
| 09/11/2025 | 1h | - | - |

Table 5.1: Hours of work spent by each author on each day on the RASD document